Fiscal Year:	FY 2006	**Task Last Updated:**	FY 01/08/2007
PI Name:	Dantzig, Jonathan A Ph.D.		
Project Title:	Numerical and Experimental Investigation of Solidification in Biological Systems		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Basics of solidification		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	61801	**Congressional District:**	15
Comments:			
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No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	4
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	0		
COI Name (Institution):	Hubel, Allison (University of Minnesota)		
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Task Description:	We combine experimental and theoretical approaches to study the role of processing parameters on the behavior of biological suspensions during freezing for cryopreservation. The experimental work involves solidification of biological suspensions (lymphocytes in aqueous solutions) on a cryomicroscope, measuring the incorporation/rejection of cells under various conditions. This work is augmented by advanced computational methods for modeling the development of microstructure during freezing, and in particular the interaction of the freezing front with cells. The objective of this research is to develop quantitative and predictive methods to improve cryopreservation protocols, to improve and enable preservation of biological cells for long term space flight. Our approach is to use numerical modeling and experimental methods to improve our understanding of the important processes which occur during freezing of cell suspensions. The important questions to address include: • Cell pushing, entrapment and engulfment by		

	the moving interface; • The role of process parameters on microstructure, and the concomitant effect on cell capture; • Redistribution of solute during freezing; • Relating all of these observations to cell survival upon thawing. This task will improve cryopreservation protocols. This will support NASA's long-term manned space flight goals, in which many biological cells will be taken in small cryopreserved samples, and grown up as needed in bioreactors. Success in this task will enable a wider variety of biological cells to be cryopreserved, and for smaller amounts to be taken in spacecraft.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The terrestrial benefits are the same as in space. Improved cryopreservation protocols will allow a wider variety of biological agents to be preserved more effectively and efficiently.
Task Progress:	Modeling Over the course of this project,we developed an adaptive level set method code to track the motion of a complex solidifying interface, and model the interaction of the dendritic front with biological cells. The model included a novel method for computation of van der Waals forces between the solid and cells. The first version of the code considered onl conduction. We then developed an implemented a Navier-Stokes fluid flow solver to work in conjunction with the solidification code, so that the drag forces on cells could be accurately computed. We also implemented a model for the expression of water from the cells as they interact with the solute-rich fluid formed by solidification. The code was verified using a suite of standard test problems, including drag forces on a fixed cylinder, and the dispersion relation for the Mullins-Sekerka instability in binary alloys. We then used the code to simulate various problems associated with freezing of biological cells in aqueous solution. This work was summarized in an article recently submitted to the Journal of Computational Physics. Experimental work We characterized microstructure for the different solutions of trehalose and saline, and growth conditions used for the experiments. We also performed a study of the post-thaw viability of cells frozen in different solutions. This work is summarized in an article recently submitted to the journal Cryobiology. Microstructure measurements Baseline microstructural measurements were performed on the solutions of interest. The primary spacing for PBS+ 10, 100 and 300 mM trehalose were measured. The primary spacing for PBS was used as a baseline or control. These studies were performed using 30 and 100 micron thick microcapillaries. For the same solution and crystal growth conditions, the primary spacings measured decreased significantly in the 30 micron capillary (compared to the 100 micron capillary). Using a smaller capillary will permit us to image cell/microstructure interactions in the same focal plane. Solidification in the larger capillaries resulted in settling of the cells into a focal plane below that of the crystal growth. Post thaw viability One of the project objectives is to relate the microstructural relations to cell survival upon thawing. To that end, lymphoblasts were frozen using a controlled rate freezer in the solutions of interest (PBS+ 10, 100 or 300 mM trehalose). The cooling rate and trehalose concentration were varied and the post thaw recovery of cells 24 h after thawing was counted. Post thaw recovery increased with increasing trehalose concentration over the range studied. The highest recovery was observed for cells frozen in 300 mM trehalose. Post thaw recovery as a function of cooling rate exhibited a characteristic inverted "U" shaped curve with optimal post thaw viability observed at 20 C/min. Period of incubation in the solution prior to freezing was also studied. A decline in post thaw recovery for cells incubated more than 30 minutes was observed. Interactions between the experimental and modeling efforts The experimental work has identified the most effective solution composition is isotonic with 300 mM trehalose addition. This composition has been shown to radically alter the solidification microstructure, changing from sharp dendrites to more highly branched structures. It is thought that this structure provides a "softer" environment for the cells.
Bibliography Type:	Description: (Last Updated: 04/03/2014)
Abstracts for Journals and Proceedings	Chang A, Dantzig JA. "Interaction of biological cells with dendritic growth." Presented at Cryobiology conference, Minneapolis, MN, July 2005. Cryobiology, July 2005. , Jul-2005
Articles in Peer-reviewed Journals	Han B, Choi JH, Dantzig JA, Bischof JC. "A quantitative analysis on latent heat of an aqueous binary mixture." Cryobiology. 2006 Feb;52(1):146-51. Epub 2005 Dec 6. PMID: 16337621 , Feb-2006
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Articles in Peer-reviewed Journals	Hubel A, Darr TB, Chang A, Dantzig J. "Cell partitioning during the directional solidification of trehalose solutions." Cryobiology. 2007 Dec;55(3):182-8. PubMed PMID: 17884036 , Dec-2007
Articles in Peer-reviewed Journals	Chang A, Dantzig JA, Darr BT, Hubel A. "Modeling the interaction of biological cells with a solidifying interface." Journal of Computational Physics, 2007 Oct 1;226(2): 1808-29. http://dx.doi.org/10.1016/j.jcp.2007.05.039 , Oct-2007